Geometric Series

Exam code: 9MA0

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Written by: Roger B
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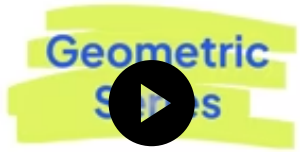

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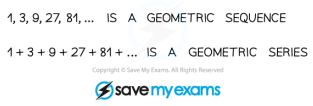
Geometric series

How do I find the sum of a geometric series?

- A **geometric series** is the sum of the terms of a **geometric sequence**



- The following formulae will let you find the sum of the first ***n*** terms of a geometric series:

$$S_n = \frac{a(1-r^n)}{1-r} \quad \text{or} \quad S_n = \frac{a(r^n-1)}{r-1}$$

  - *a* is the first term
  - *r* is the common ratio

- The one on the left is more convenient if ***r* < 1**, the one on the right is more convenient if ***r* > 1**
- The ***a*** and the ***r*** in those formulae are exactly the same as the ones used with **geometric sequences**

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- Learn this proof of the geometric series formula – you can be asked to give it in the exam:
  - Write out the sum once
  - Write out the sum again but multiply each term by *r*
  - Subtract the second sum from the first
    - All the terms except two should cancel out
  - Factorise and rearrange to make *S* the subject

THE SUM OF THE FIRST n TERMS IS

$$S_n = a + ar + ar^2 + ar^3 + \ldots + ar^{n-2} + ar^{n-1} \qquad (1)$$

MULTIPLYING BY r GIVES

$$rS_n = ar + ar^2 + ar^3 + ar^4 + \ldots + ar^{n-1} + ar^n \qquad (2)$$

SUBTRACT (2) FROM (1) AND REARRANGE

$$S_n - rS_n = a - ar^n$$

$$S_n(1-r) = a(1-r^n)$$

$$S_n = \frac{a(1-r^n)}{(1-r)}$$

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What is the sum to infinity of a geometric series?

- If (and only if!) $|r| < 1$, then the geometric series **converges** to a finite value given by the formula

$$S_\infty = \frac{a}{1-r}$$

- S_∞ is known as the **sum to infinity**
- If $|r| \geq 1$ the geometric series is **divergent** and the sum to infinity does not exist

Examiner Tips and Tricks

- The geometric series formulae are in the formulae booklet – you don't need to memorise them
- You will sometimes need to use logarithms to answer geometric series questions (see **Exponential Equations**)

Worked Example

$10 + 9 + 8.1 + \ldots$ to n terms exceeds 99.

b) What is the sum to infinity of the series?

a) THIS IS A GEOMETRIC SERIES WITH

$a = 10 \quad r = 0.9$

WE WANT

$$S_n = \frac{10(1-0.9^n)}{1-0.9} > 99$$

$$100(1-0.9^n) > 99$$

$$100 - 100(0.9^n) > 99$$

$$100(0.9^n) < 1$$

$$0.9^n < 0.01$$

$$\log(0.9^n) < \log 0.01$$

TAKE LOGARITHMS

$$n \log 0.9 < \log 0.01$$

$$n > \frac{\log 0.01}{\log 0.9}$$

log 0.9 IS NEGATIVE, SO INEQUALITY SIGN FLIPS

$$n > 43.7\ldots$$

$$n = 44$$

n HAS TO BE AN INTEGER

b) WITH $a = 10$ AND $r = 0.9$

$$S_\infty = \frac{10}{1-0.9} = 100$$

THE SUM TO INFINITY IS 100

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